

# Project Development Phase

## Phase 5 of 8 - Project Documentation

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*Project: OptiCrop: Smart Agricultural Production Optimization Engine*

### Project Folder Structure

- Opti\_Crop/Dataset/Crop\_recommendation.csv - the raw dataset.
- Opti\_Crop/Training/OptiCrop\_Model\_Building.ipynb - the full training notebook.
- Opti\_Crop/Flask/app.py - the Flask backend.
- Opti\_Crop/Flask/model.pkl - the serialized trained Logistic Regression model.
- Opti\_Crop/Flask/templates/ - home.html, about.html, findyourcrop.html.

### Development Summary by Epic

#### Data Collection and Analysis

The Crop\_recommendation.csv dataset (2,200 rows, 22 crop labels, 100 samples each) was loaded and explored using `df.shape`, `df.info()`, and `df.describe()`. Univariate analysis examined the distribution of each soil/climate feature; bivariate analysis visualized humidity against crop label; multivariate analysis used a correlation heatmap and pair plots across N, P, K, and temperature.

#### Data Pre-Processing

No null values were found in the dataset. Outliers in the Phosphorous feature were identified via boxplot and removed using the IQR method (values outside  $Q1 - 1.5 \cdot IQR$  to  $Q3 + 1.5 \cdot IQR$ ). Seasonal crop subsets (summer, winter, rainy) were extracted based on temperature, humidity, and rainfall thresholds. The cleaned dataset was split 80/20 into training and test sets.

#### Model Building

K-Means clustering ( $k=4$ , `k-means++` initialization) was applied to the full feature set to explore natural groupings of crops by environmental similarity, guided by the Elbow Method. A Logistic Regression classifier was then trained on the 80% training split and evaluated on the remaining 20% test split using a full classification report (precision, recall, F1-score per crop) and overall accuracy.

#### Application Building

The Flask backend loads `model.pkl` at startup and exposes three page routes (`/`, `/about`, `/findyourcrop`) plus a `/predict` POST route that reads the seven form values, builds a NumPy feature array in the fixed order [N, P, K, temperature, humidity, ph, rainfall], and renders the recommended crop back onto the Find Your Crop page.